Ben Bubnick

Master Career History

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Career Overview

Comprehensive working record spanning healthcare data consulting, data engineering, analytics, software development, scientific computing, entrepreneurship, higher education, academic research, farm operations, and community service. Combines hands-on delivery with technical leadership, documentation, teaching, stakeholder engagement, business administration, and quality-focused practices. Maintained as a source for targeted resumes, biographies, applications, and future career pivots.

Skills and Domains

Data Engineering and Software: SQL, Python, Ruby, Snowflake, dbt, SQLMesh, Spark, Hadoop/HDFS, Hive, Impala, Kafka, PostgreSQL, Oracle, C#, VB.NET, C++, FORTRAN, ETL, ELT, data warehouses, data lakes
Cloud and Delivery: AWS, Azure, Docker, Git/GitHub, GitHub Actions, Jenkins, Azure DevOps, Jira, Confluence, CI/CD, technical documentation, mentoring
Healthcare and Analytics: Clinical, payer, eligibility, revenue-cycle, and financial data; HL7, FHIR, EDI/837, HEDIS, EHR/EMR, HIPAA, HITRUST, pandas, Jupyter, Power BI, Tableau, statistical modeling
Science and Education: Spectroscopy, photometry, optical measurement, radiative-transfer modeling, Fourier analysis, Monte Carlo methods, numerical integration and interpolation, physics and astronomy instruction, curriculum development, research publication
Business Operations: Bookkeeping, accounting, tax compliance, marketing, promotional-material development, client acquisition
Earlier Scientific and Creative Tools: R, PERL, IDL, IRAF, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP, SAS/GRAPH, AutoCAD, Adobe Illustrator, Adobe Photoshop, GIMP, LaTeX

Professional Experience

HHAeXchange Oct. 2025 - Jun. 2026
Healthcare Data Integration Consultant - Contract via Veersa Technologies
Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox
Led healthcare source analysis, data-quality investigation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Structured target models using FHIR-like concepts where appropriate to improve consistency and extensibility for downstream integration and analytics.
- Defined surrogate-key, natural-key, and crosswalk strategies supporting stable identity resolution across related source systems.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
- Established reusable patterns for onboarding additional sources and extending integration workflows.

Adonis Mar. 2025 - Jul. 2025
Healthcare Revenue Cycle Data Consultant - Contract via Health Data Movers
Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker, GitHub Actions
Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.

- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Diagnosed undocumented source-key versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.
- Built a reusable SQLMesh/Snowflake ingestion and transformation framework using templated models, data contracts, CI checks, and promotion gates.
- Added schema-drift alerts, parity checks, reconciliation logs, deterministic rerun procedures, and incident-handling guidance for UNIX-like environments.
- Packaged a lightweight operations forecasting and alerting pattern as a tested Python and Docker service with GitHub Actions deployment automation.
- Standardized transformation logic across clients and translated business requirements into technical specifications to improve delivery predictability.

GrowData Analytics Sep. 2024 - Oct. 2025
Independent Data Automation and Analytics Consultant
Technologies: Python, Ruby, RSpec, Tesseract OCR, LLM APIs, Docker, AWS, GitHub Actions

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.
- Productized Tesseract OCR and rule-based post-processing as a small Python/Ruby service with containerized deployments and versioned configuration.
- Added schema checks, cross-field validation, reconciliation, idempotent reruns, fixtures, golden-file tests, linting, and release automation.
- Documented runbooks and failure modes covering timeouts, low-confidence OCR, and retries.
- Built a lightweight analytics layer tracking usage, error categories, latency, and throughput to guide feature tuning and cost control.
- Managed infrastructure as code for a minimal AWS footprint using S3, Lambda or scheduled jobs, and CloudWatch alerts through Bash/Zsh and Makefile workflows.
- Managed the business's bookkeeping, accounting, and tax obligations.
- Led marketing and client acquisition, including creating promotional materials for the business.

Arkos Health Jun. 2023 - Jan. 2025
Senior Data Analyst
Technologies: AWS S3, Athena, Redshift, Glue, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets; CI/CD and targeted model refactoring reduced deployment time by approximately 50% and improved query performance.
- Integrated Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting; recorded notes attributed a 30% increase in stakeholder satisfaction and a 40% reduction in new-team-member onboarding time to this work.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.
- Mentored engineers on code review, testing, and pipeline design and authored internal delivery standards.
- Implemented Jira and GitHub workflows; recorded notes attributed a 25% improvement in task tracking and team coordination within three months to the change.
- Expanded cross-departmental collaboration; recorded notes attributed a 15% increase in data-driven decision-making within six months to this work.
- Mentored junior team members; recorded notes attributed a 30% improvement in their performance metrics within six months to this support.
- Led goal-setting sessions; recorded notes attributed a 30% improvement in goal alignment and tracking accuracy during the first quarter.
- Initiated stakeholder-feedback sessions; recorded notes attributed a 25% improvement in deliverable relevance and accuracy within three months.

Merative Jul. 2022 - Mar. 2023
Senior Data Scientist
Technologies: Azure Data Factory, Blob Storage, Functions, SQL Database, Synapse Analytics, Python, SQL, Azure

DevOps

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition reconciliation, schema compatibility, null handling, duplicates, field-level comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity.
- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.
- Designed and implemented automated ETL testing and regression suites that detected schema drift and transformation errors early in the migration lifecycle.
- Validated end-to-end Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations during the transition from HDFS to Azure.
- Collaborated with multidisciplinary engineering teams to refine architectural designs and preserve workflow continuity and data fidelity.
- Documented test plans, platform-migration procedures, and onboarding guidance for future quality-assurance engineers.
- Enforced data-governance and security protocols supporting audit-risk reduction and HITRUST readiness.

IBM Jun. 2015 - Jul. 2022

Advanced through five roles delivering healthcare analytics, data architecture, client integration, data-quality improvement, and large-scale migration validation.

Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, PostgreSQL, Docker, Jenkins, Tableau, HL7

- Supported employee onboarding from 2016 to 2018, helping new colleagues learn team practices, systems, and technical workflows.
- Served on tiger teams supporting the HDFS-to-Azure data-platform upgrade and participated on the data-governance team.
- Participated in internal coding jams, hackathons, and cross-team delivery initiatives.
- Played a central role in the REACH cross-team training initiative, teaching SQL fundamentals to nontechnical colleagues and ETL, architecture, data-science, and predictive-modeling practices to technical colleagues.

Senior Data Scientist Nov. 2021 - Jul. 2022

- Served as a lead member of Provider Analytics, integrating disparate and centralized sources into data marts and supporting data mining, statistical-model development, and advanced analytics.
- Worked with senior leaders and colleagues to identify opportunities, define objectives, and formulate analytics strategies that produced measurable impact.
- Served as technical lead across electronic health records, insurance claims, scheduling and financial data, social determinants of health, and other relevant sources.
- Project note: HDP Data Migration.
- Project note: Phytel Support.
- Project note: HIE P3N.
- Project note: Explorys Decommissioning.
- Project note: GP Support for CCF.
- Project note: Phytel Configuration Training.
- Project note: Jira Migration Validation.
- Project note: Azure REACH.
- Led delivery of a tailored analytics and reporting project using client administrative data.
- Developed data-migration validation analysis spanning more than 150 billion patient records.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.

Senior Data Architect Dec. 2020 - Nov. 2021

- Served as a senior member of Data Onboarding & Governance, integrating healthcare data, enriching metadata, and configuring data models, analytics, reports, and visualizations.
- Served as a subject-matter expert to project data architects and delivery, analytics, and integration teams; directed configuration of the data-onboarding platform and supporting infrastructure.
- Led a cross-functional team as the clinical data analyst for the 2020/2021 measure-updates project.
- Created a machine-learning COVID-19 tool that reduced the time required to identify the need for hospitalization by up to two weeks.
- Developed a topic-modeling project using nearly six million semi-structured provider notes.
- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept that was being evaluated as a potential new offering.
- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
- Led a HIPAA patient-deidentification project drawing from more than 100 data sources for content analytics and

measures.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.
- Served in an advisory capacity for development of a new data platform.
- Closed project-documentation gaps by organizing material from more than 100 past projects.

Data Architect Nov. 2017 - Dec. 2020
- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.
- Built the data-onboarding platform and associated infrastructure, remediated UAT data issues, and reported on data health during onboarding.
- Earned AI certification to peer-review patient-matching algorithm tuning and reviewed more than 50% of projects during the final two quarters of the period recorded in the original notes.
- Curated 25 pages of AI-workflow documentation and mentored new hires in the AI-tuning process.
- Spearheaded a new-hire mentoring program associated with an approximately 4,500-hour increase in total billable time.
- Received the Lab Services Award for coordinating internal project teams to deliver client needs on time.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.34%.

Software Developer, Data Platform Sep. 2016 - Nov. 2017
- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating $30,000 in annual vendor costs.
- Reduced time spent on knowledge transfer by approximately 80% by rewriting and reorganizing project documentation.
- Reduced cost per hire by $5,210 by redeveloping the employee education program.
- Received the Eminence and Excellence Award in 2017 for an analytics project.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.17%.

Data Scientist Nov. 2015 - Sep. 2016
- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Served as principal integration specialist on more than 60 projects for 20 clients across the broader 2015–2018 data-ingestion period.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.
- Built a Python, Ruby, and SQL extraction tool that reduced total client-integration effort by 28%, measured through eliminated story points and lower average engineering hours per implementation.
- Eliminated an average of three weeks from employee onboarding.

Explorys, an IBM Company Jun. 2015 - Nov. 2015
Data Science Consultant - Contract via Randstad USA
Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins
Supported clinical-data integration and population analytics across more than 10 client data warehouses. IBM acquired Explorys in 2015; the overlap with the IBM tenure reflects the acquisition and transition rather than two unrelated concurrent roles.
- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.
- Identified and extracted data from heterogeneous client systems; led technical requirements gathering, discovery, documentation, and implementation; and coordinated acquisition, installation, and configuration of VPN hardware supporting client integrations.
- Developed a cron-based control for defining blackout periods around client data pulls.

AmTrust Financial Services Mar. 2014 - Jun. 2015
Application Development Consultant - Contract via TEKsystems
Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL, R
Developed ISO-based web forms, multi-tier MVC components, Web APIs, and user interfaces for commercial insurance policy software.
- Designed, developed, tested, and maintained full-stack commercial-insurance applications spanning VB.NET and ASP.NET interfaces, multi-tier components, Web APIs, and SQL data services.
- Implemented a new line of business in VB.NET and Microsoft SQL Server, served as principal developer for a new product offering, and contributed to two major software-architecture initiatives in 2015.
- Evaluated the advantages, risks, and implementation costs of requested features and recommended maintainable

solutions grounded in object-oriented design principles.
- Served as the working liaison between business analysts and developers, translating requirements, reporting progress, and helping prioritize delivery needs.
- Coordinated developer schedules and troubleshot production problems across multiple application environments.
- Created a unit- and integration-test suite for multi-tier MVC components, improving regression coverage and establishing stronger team testing practices.
- Developed more than 65 ISO rating-based web forms and analyzed page timings in R to identify optimization targets and improve application performance.
- Refactored and maintained the existing VB.NET insurance-rating codebase to improve readability, performance, and long-term maintainability.
- Supported the technical integration of acquired businesses into the commercial policy platform.

PPG Industries Apr. 2012 - Aug. 2013
Color Scientist
Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods
Developed software and radiative-transfer models for automotive paint formulation, spectroscopic profile analysis, and technology-risk assessment.
- Improved a color-matching algorithm by 40% using numerical integration methods.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.
- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.
- Led cross-functional color-science and marketing initiatives from concept through on-time, in-full delivery for the Automotive Refinish business.
- Served as business owner for color data-management software, directing its global development, launch, training, adoption, and application.
- Managed relationships with third-party equipment and software suppliers, acting as the primary contact for technical coordination and issue resolution.
- Managed a portfolio of color-related intellectual property and supported the translation of research into operational tools and practices.
- Served as the global liaison among Color, Technical, Operations, and IT teams, translating color-science requirements into consistent systems, workflows, and communications.
- Designed and standardized color workflows and best practices for internal operations and external customers, leading the change-management efforts required for adoption.
- Evaluated color processes, instrumentation, and data flows to identify technical and operational risks and develop practical mitigation strategies.
- Developed statistical models from established theory, experimental data, and novel approaches to improve color measurement, matching, and process decisions.
- Championed a more science-centered organizational approach to color by introducing analytical methods, shared standards, and evidence-based decision-making.
- Informed marketing strategy and communications by connecting customer needs with technical capabilities and broad operational insight.

Teaching and Academic Research

Lorain County Community College Aug. 2016 - May 2019
Adjunct Professor - Taught the laboratory component of algebra-based General Physics I (then PHYC 151) and the three-credit ASTY 152 Solar System Astronomy lecture course.
- Guided students through laboratory work in mechanics, energy, momentum, rotation, and thermal physics, emphasizing measurement uncertainty, quantitative analysis, and scientific reporting.
- Developed and taught online astronomy instruction covering astronomical observation, planetary-system formation, terrestrial and Jovian planets, small bodies, and space exploration.
- Exceeded institutional student-evaluation benchmarks by an average of 16% in 2017 and 17% in 2018.
- Reduced reported student confusion by 28% by reorganizing online instructional materials and reducing the need for clarification during office hours.
- Developed and published more than 70 hours of recorded lectures and laboratory content to support flexible, on-demand learning.
- Used Just-in-Time Teaching strategies with underperforming students; recorded notes attributed an average 18% end-of-term grade improvement to this work.

Miami University Aug. 2010 - Jun. 2011
Adjunct Professor - Developed and taught three-credit general-education courses PHY 101 Physics and Society and PHY 111 Astronomy and Space Physics for more than 100 students.

- Designed a concept-first applied-physics curriculum for non-science majors, translating advanced technical subjects into accessible, decision-ready frameworks without relying on calculus.
- Taught students to use order-of-magnitude estimates, energy density, power, conservation laws, and comparative risk to evaluate claims involving transportation fuels, electricity generation, climate, and emerging technologies.
- Connected nuclear physics and radioactivity with power generation, medical imaging, radiation risk, and nuclear security; linked gravity, waves, and the electromagnetic spectrum with satellites, GPS, radar, communications, and remote sensing.
- Structured lessons around consequential real-world questions, teaching students to distinguish scientific constraints from policy and value judgments and to communicate evidence-based conclusions to nontechnical audiences.
- Taught astronomy topics spanning planetary systems, stellar evolution, galaxies, cosmology, and the scientific search for extraterrestrial life.

Cincinnati State Technical and Community College Feb. 2011 - Jun. 2011
Adjunct Professor - Developed and delivered the integrated Technical Physics I–III lecture and laboratory sequence for technical-program students with varied mathematics preparation.

- Delivered two weekly lecture hours and three laboratory hours per course across applied mechanics, thermodynamics, electricity and circuits, waves, and optics.
- Wrote laboratory instructions and coordinated measurement, data-analysis, and technical-report work for five student teams.

University of Cincinnati Oct. 2008 - Jun. 2010
Graduate Teaching Assistant - Led approximately 60 laboratory meetings for more than 100 engineering, science, and preprofessional students in the algebra/trigonometry-based College Physics sequence (PHYS 111–113) and calculus-based General Physics sequence (PHYS 211–213).

- Taught experimental design, measurement uncertainty, data acquisition, graphical analysis, and technical reporting using PASCO DataStudio and Microsoft Excel.
- Served as teaching assistant and grader for the corresponding College Physics lecture sequence (PHYS 101–103) and General Physics lecture sequence (PHYS 201–203).
- Supported Astronomy and the Natural Universe lectures (PHYS 121–122) and astronomy laboratories (PHYS 125–127) through grading and instructional assistance.
- Designed interactive laboratory instruction using Peer Instruction and Interactive Lecture Demonstrations to identify and address student misconceptions in real time.

University of Cincinnati Oct. 2009 - Dec. 2009
Engineering Teaching Assistant - Prepared and delivered Mathematica instruction for Numerical Methods for Engineering Design, an upper-division aerospace-engineering course.

- Developed step-by-step demonstrations applying symbolic and numerical computing to interpolation and approximation, numerical integration, nonlinear equation solving, systems of equations, and ordinary differential equations.
- Connected computational methods with aerospace analysis and design problems, showing students how to translate mathematical formulations into executable workflows and interpret the resulting output.
- Reinforced numerical accuracy, error, stability, and method selection so students could evaluate the credibility of computed engineering results.

University of Cincinnati Summer 2006
Supplemental Instruction Leader - Supported an intensive summer program in which academically advanced incoming students completed the full calculus-based introductory physics sequence, PHYS 201–203, before beginning the regular academic year.

- Delivered focused problem-solving lectures between formal class meetings, working representative problems step by step and demonstrating efficient solution strategies.
- Helped students connect concepts across the compressed three-course sequence through diagrams, dimensional analysis, estimation, and physical interpretation of calculated results.
- Led interactive sessions for groups of up to 12 students, addressing misconceptions and providing timely reinforcement between accelerated course meetings.

University of Cincinnati Jan. 2006 - May 2006
Undergraduate Teaching Assistant - Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.

University of Cincinnati Learning Communities Program Jun. 2006 - Aug. 2008
Peer Leader - Facilitated a learning community for approximately 25 first-year students co-enrolled in multiple courses, supporting their academic, social, and professional transition to university life.

- Prepared and facilitated twice-weekly meetings on course assignments, study strategies, time management, academic expectations, campus resources, and first-year transition needs.
- Provided individual and small-group mentoring, helping students form study networks, navigate university systems,

and connect with faculty, advisers, and academic-support services.
- Planned community-building activities and coordinated with faculty, advisers, program staff, and campus organizations to strengthen belonging, peer accountability, and academic engagement.

Private Tutoring Jun. 2006 - Aug. 2008
Private Tutor - Provided sustained one-on-one instruction to a student completing the calculus-based introductory physics sequence, PHYS 201–203.
- Adapted short lectures, demonstrations, and representative worked problems to the student's individual learning needs and progress across the three-course sequence.
- Taught a repeatable problem-solving framework built around diagrams, governing physical principles, dimensional analysis, estimation, and interpretation of calculated results.
- Diagnosed misconceptions through guided questioning and targeted practice, reinforcing difficult concepts with step-by-step explanations and alternative solution strategies.

Across these institutions, used interactive, peer-led, and just-in-time teaching methods.

University of Cincinnati Sep. 2006 - Sep. 2010
Research Assistant, Department of Physics - Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.
- Reduced and analyzed 0.8–2.4 micrometer, $R \sim 2000$ SpeX spectra and measured diagnostic H I, He I, He II, C IV, N III, and CO spectral features across nine candidate cluster members.
- Co-developed an astrometrically calibrated catalog of 130 sources by registering SpeX acquisition images to 2MASS; compiled available JHK photometry and used DAOPHOT to separate blended sources in the crowded field.
- Helped classify the cluster's massive stellar population, identifying an O5 V dwarf, several early-B stars, and a massive young stellar object exhibiting CO band-head emission from dense circumstellar material.
- Combined spectral classifications with 2MASS photometry to estimate cluster extinction ($A_V \sim 21$–23), distance (1.0–1.75 kpc), age of no more than a few million years, and mass of approximately 10^3 solar masses.
- Cross-matched infrared and radio-source positions and identified candidate near-infrared counterparts for three deeply embedded radio sources associated with early star formation.
- Published the resulting 130-source astrometric and photometric catalog through the international VizieR astronomical data service.
- Acquired follow-up L-band spectra of Galactic OB stars during three consecutive nights at NASA's IRTF, replicating earlier SpeX measurements and validating the use of Brα and Pfγ diagnostics to characterize stellar-wind mass loss and clumping.
- Estimated stellar radial velocities by measuring pixel-space shifts of CO absorption features against normalized standard-star spectra.
- Performed astronomical spectroscopy, photometry, classification, calibration, signal analysis, and noise reduction using IRAF, IDL, SpeXtool, Mathematica, and Perl.

Ohio University Oct. 2003 - May 2004
Undergraduate Student Researcher, Printmaking - Used a competitive undergraduate research grant to adapt a wintergreen-oil toner-transfer process from lithography to intaglio photo-etch printing.
- Developed and tested a complete copper- and zinc-plate workflow spanning surface preparation, asphaltum grounding, high-contrast toner transfer, burnishing, clearing, aquatint, and staged etching.
- Replaced xylene- and acetone-based photo-transfer steps with methyl salicylate, eliminated the need for specialized exposure equipment, and demonstrated a process using less than $20 in materials.
- Characterized critical process variables and failure modes, including ground thickness and cure time, solvent saturation, image resolution, transfer pressure, evaporation time, and temperature control during aquatint fusion.

Other Experience

Doodlebug Farms Sep. 2024 - Present
Family Farm Operator - Runs a family farm with 80 trees and manages its ongoing expansion.
Lincoln Park Pub Aug. 2013 - Mar. 2014
Cook - Worked in the restaurant kitchen.
University of Cincinnati Geology Math Physics Library Mar. 2006 - Sep. 2008
Student Librarian - Assisted patrons with research and reference needs, maintained Library of Congress-classified materials, conducted inventory, and located missing items.
FranklinCovey Jun. 2005 - Apr. 2006
Productivity Consultant and Sales Associate - Advised customers on planning products and workshops, used detailed product knowledge to meet client needs, and supported cash and credit transactions.
Retail Food Service and Manual Work 1995 - 2010
Various roles - Worked as a Target cashier; Dillard's sales associate; Aladdin's Eatery dishwasher; Cravings Cafe and Take 5 Coffee Company barista; Pickle Bill's Lobster House cook; and Pier W dishwasher and cook. Other early work included retail sales, coffee-shop management, telemarketing, industrial labor, FedEx loading, interior painting, cement work, yard work, swim instruction, and grocery and restaurant service.

The customer-facing and operations roles included opening and closing, training coworkers, ordering and receiving inventory, stocking and heavy lifting, handling cash and credit transactions, preparing food and espresso drinks, and resolving customer concerns during high-volume service.

Volunteer and Community Experience

HSEA Feb. 2025 - May 2025
ETL Brigade Volunteer - Built a reusable PostgreSQL model for community-health analytics covering claims, visits, and eligibility.
CSforAll Oct. 2018 - May 2019
Instructor - Supported computer-science instruction for high-school students, with emphasis on learners without reliable home computer or Internet access.
HIT in the CLE Mar. 2017 - Mar. 2019
Coach - Used a data-science competition to introduce students to health-information-technology careers and support the regional talent pipeline.
Code for America Jan. 2015 - Jun. 2015
ETL Guild Volunteer - Applied data-science tools to improve and expand public-record access.
Heights Observer, Cleveland Heights, Ohio Feb. 2012 - May 2014
Volunteer Editor - Edited community-newspaper content in support of local civic engagement and an independent free press.
The News Record, University of Cincinnati Feb. 2008 - Sep. 2010
Editorial Contributor - Wrote and edited content for the university's student newspaper.
TCP World Academy Sep. 2008 - May 2009
Tutor - Helped students connect classroom learning with participatory society-simulation activities.
Project Astro at Ethel M Taylor Academy 2008
Volunteer Astronomer - Supported a national program connecting astronomers with educators to strengthen astronomy and physical-science instruction.
Reach Out on Campus at Ohio University Sep. 2000 - Jun. 2004
Volunteer - Participated in campus-based community service.

Leadership and Professional Affiliations

Detroit Color Council 2012 - 2013
Member
American Institute of Aeronautics and Astronautics Cincinnati Student Chapter 2009 - 2011
Vice President in 2010; participated in Students for the Exploration and Development of Space in 2009.
University of Cincinnati Astronomy Club 2008 - 2010
Founding member in 2008, treasurer from 2008 to 2009, and secretary from 2009 to 2010.
Society of Physics Students Cincinnati Chapter 2005 - 2008
Secretary from 2005 to 2006 and treasurer from 2006 to 2007.
Sigma Theta Epsilon Beta Gamma Chapter 2005 - 2008
Held secretary, treasurer, chaplain, and publicity responsibilities.
Additional Academic Activities 2006 - 2010
Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, the University of Cincinnati Mountaineering Club, and University of Cincinnati physics departmental seminars from 2008 to 2010.
Athens Wargamers Jan. 2001 - Jun. 2004
Member; served as treasurer during the 2002–2003 academic year.

Creative and Visual Arts

Zygote Press 2011 - 2014
Exhibiting Artist - Participated in printmaking exhibitions organized by the studio.
Senior B.F.A. Exhibition, Jennings House Gallery May 2004
Exhibiting Artist - Presented senior printmaking work at Ohio University.
Student Printmakers Exhibition and Print Exchange Feb. 2004
Organizer - Organized an exhibition and print exchange for student printmakers.
Lacuna Gallery, Logan, Ohio 2004
Exhibiting Artist - Displayed original artwork in a local gallery exhibition.
Ohio University School of Art Commission 2003 - 2004
Commissioned Artist - Created an original print for the School of Art president; the exact commission date and recipient name are not recorded.
Ohio University Visiting-Artist Print Projects 2000 - 2004
Student Assistant - Assisted visiting artists Debra Fisher and Dan Murphy with two print projects.
Lakewood Hospital, Lakewood, Ohio 1999
Exhibiting Artist - Displayed artwork at the hospital.

Selected Independent Projects

iphone-cam-lite 2025 - 2026
Native Swift/SwiftUI macOS menu-bar app that selects and verifies iPhone Continuity Camera devices in Zoom and Microsoft Teams; includes permission-aware onboarding, diagnostics, persistent settings, automated tests, CI, and release packaging.

farm-invoice-parser 2025
Ruby, RubyLLM, OCR, and model APIs; converts handwritten farm invoices into structured transaction records.

cfd_snow_fence 2025
Python applied-modeling project combining measured wind data, Navier-Stokes pressure calculations, and XGBoost classification to identify potential snow-drift locations.

healthcare-dbt-demo 2025
Runnable Data Vault 2.0 demonstration for healthcare claims data using dbt and DuckDB/PostgreSQL.

claims_case_study 2023
Healthcare staffing-demand analysis using data-quality assessment, Parquet transformation, exploratory analysis, and ARIMA forecasting; delivered a written report and stakeholder presentation.

HL7-ingestion 2023
Java and Docker healthcare-interoperability project converting HL7 messages into longitudinal patient records.

quality_measures_engine 2023
Python project for healthcare quality-measure processing and evaluation workflows.

DETL-design-challenge 2023
Health-data transformation and loading design demonstrating ETL automation and validation.

zams_cluster_candidacy 2019 - 2020
Python scientific-analysis pipeline using zero-age main-sequence and isochrone comparisons to evaluate stellar-cluster candidacy.

gamma_rays 2019
Python and Jupyter experiments in gamma-ray classification, energy reconstruction, and deep learning, with disseminated analysis outputs.

Star-Chart 2015
JavaScript web application displaying the night sky for a selected location and date using stellar catalog data.

Publications and Research Outputs

Journal Articles and Data Catalogs

Bubnick, B. (2015). *AI paint formula prediction from spectroscopic measurements* (Version 1.2) [Interactive web application]. shinyapps.io.
Bubnick, B. F. (2010). *Massive stellar clusters in the disk of the Milky Way Galaxy* [Master's thesis, University of Cincinnati]. OhioLINK Electronic Theses and Dissertations Center. OhioLINK record
Hanson, M. M., & Bubnick, B. F. (2008). *JHK photometry in cluster [BDS2003] 107* [Data set]. VizieR Online Data Catalog (Catalog J/PASP/120/150). `https://doi.org/10.26093/cds/vizier.61200150`
Hanson, M. M., & Bubnick, B. F. (2008). Near-infrared spectroscopy of candidate members of the Galactic cluster [BDS2003] 107. *Publications of the Astronomical Society of the Pacific, 120*(864), 150–159. https://doi.org/10.1086/527403
Bubnick, Ben. "A Novel Method for Low-Toxic Photo-Etch Printing." *MAPC Journal*, vol. 11, no. 2, Fall/Winter 2003, pp. 13-14.

Applied Research and Technical Writing

Bubnick, B. (2025). "Reproducible Postgres load of Tuva seed datasets." Internal applied research series, GrowData Analytics/HSEA. Document tracking no. 20250401-A. Internally published by HSEA, Aug. 2025. Author's version, Apr. 1, 2025.
Ben Bubnick. (2020). "Surgical Scheduling Optimization and Procedure Duration Prediction." Internal project report, IBM. Document tracking no. RC25712.
Ben Bubnick. (2017). "Process and Requirements for CCD Data Integration." Internal white paper, IBM. Document tracking no. RC25043.
Ben Bubnick. (2013). "Characterization of Surface Coatings Using RT Models for Color Matching." Internal technical memorandum, PPG Industries. Ref. 1284 - February 2013.
Ben Bubnick. (2010). "The B Stars of Westerlund 1." Internal applied research series, University of Cincinnati MASSCLEAN Project. `https://doi.org/10.1088/0004-6256/138/6/1724`

Speaking and Knowledge Sharing

AI Hackathon: Surgery Scheduling Optimization Dec. 2020
Presented an approach using historical utilization and current wait-list information to predict surgery duration.
REACH Initiative: IBM Watson Care Insights 2019
Presented the use of patient records and real-time HL7 feeds to integrate insights into clinical workflows.
Content and Data Analytics Learning Exchange 2018
Introduced machine-learning implementation using real-world health-data analytics.
Innovation Research Interchange Fall Networks Conference Sep. 2018
Presented natural-language processing for extracting insight from unstructured data lakes.
Cross-Team Training Series 2016
Led a seven-part series on Hadoop ingestion and data-lake concepts.
M.S. Physics Thesis Defense, University of Cincinnati Jun. 2010
Defended a thesis presenting spectroscopic and photometric analysis of two open clusters in the Galactic plane.
2MASS Research Poster Session 2008
Presented a poster on astronomical research using Two Micron All Sky Survey (2MASS) data.

Professional Recognition

Special Recognition, Health Data Movers - client delivery 2025
Arkos Health BI Star - Salesforce Trailhead Expeditioner 2023
Manager Recognition for Transition Leadership, Merative - systems migration 2022
Gold Champion, IBM 2018 - 2022
Lab Services Award, IBM - cross-team delivery coordination 2018
Eminence and Excellence Award, IBM - data-curation tool development 2017
Color Lab Automation Award, PPG Industries - expanded reporting 2012
Mary J. Hanna Research Fellowship, University of Cincinnati 2010
Academic Scholarships and Honors 2000 - 2011
University Graduate Scholarships for physics study from 2008 to 2010 and aerospace-engineering study in 2011; University of Cincinnati Dean's List from 2005 to 2007; Ohio University Provost's Undergraduate Research Grant in 2003 and Dean's List from 2002 to 2004; Ohio Proficiency Scholarship in 2000.

Education

Graduate study, Aerospace Engineering, University of Cincinnati, 2011 - Computational Fluid Dynamics; degree not completed
M.S. Physics, Astrophysics concentration, University of Cincinnati, Dec. 2010 - thesis option; GPA 3.3/4.0

B.F.A. Printmaking, Ohio University, Aug. 2004 - major GPA 3.6/4.0
Additional coursework: Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning; astrobiology; critical thinking

Professional Development and Credentials

Fundamentals of Visualization with Tableau - University of California, Davis Jun. 2023
Credential ID M3VVMS4BTXWU.
IBM Data Science Professional Certificate 2022
Certificate of Specialization, Data Science - Johns Hopkins University 2015
IBM Digital Credentials and Coursework Nov. 2017 - Feb. 2022
Completed 43 distinct IBM credentials spanning data science, machine learning, big data, cloud, product management, design thinking, security, blockchain, and quantum computing.

- **Nov. 2017:** IBM Agile Explorer.
- **Feb. 2018:** IBM Blockchain Essentials.
- **May 2018:** IBM Cloud Private Infrastructure and Architecture; Hadoop Data Access - Level 1.
- **Jun. 2018:** Product Manager - Define, Build & Deliver; Manage the Business; Product Manager - Scaling the Business.
- **Aug. 2018:** Cognitive Practitioner.
- **Feb. 2020:** IBM Quantum Conversations.
- **Sep. 2020:** AgileTA Explorer; Think Like a Hacker.
- **Oct. 2020:** Watson Visual Recognition; Enterprise Design Thinking Practitioner; IBM Cloud Private - Foundation Technology; Watson and Cloud Foundations; Enterprise Design Thinking Co-Creator.
- **Jan. 2021:** Cloud Service Management and Operations Explorer.
- **Jun. 2021:** Data Science Foundations - Level 1.
- **Jan. 2022:** Data Science Methodologies; Python for Data Science; Hadoop Programming - Level 1; Hadoop Foundations - Level 1; Cloud Core; Data Analysis Using Python; Simplifying Data Pipelines with Apache Kafka; Deep Learning Essentials; Spark - Level 1; Data Science with Scala; Applied Data Science with Python - Level 2; Data Visualization Using Python; Machine Learning with Python - Level 1; Data Science Tools; IBM Blockchain Essentials V2; Introduction to Machine Learning with Sound; Big Data Foundations - Level 2; Big Data Foundations - Level 1; Machine Learning with R - Level 1; Data Visualization with R; Deep Learning using TensorFlow; Deep Learning; Accelerated Deep Learning with GPU; Data Science Foundations - Level 2 (V2).
- **Feb. 2022:** IBM AI Associate Data Scientist.

Salesforce Trailhead Coursework Completed at Arkos Health 2023 - 2025
Reached Trailblazer Mountaineer rank with 18,725 points and 40 badges.

- **Tableau and data literacy:** Tableau Prep Builder Basics; Tableau Blueprint Basics; Tableau Visualization Formatting on the Web; Tableau AI and Analytics: Quick Look; Data Sharing and Collaboration in Tableau Cloud; Data Visualization in Tableau Cloud; Tableau Cloud Basics; Data Presentation and Publication in Tableau Desktop; Data Filtering and Sorting in Tableau Desktop; Data Analysis in Tableau Desktop; Data Connection in Tableau Desktop; Tableau Data Management: Quick Look; Variation for Data Comparisons; Data Distributions; Aggregation and Granularity; Variables and Field Types; Well-Structured Data; Data Literacy Basics; The Tableau Desktop Workspace; The Tableau Workflow.
- **Salesforce, analytics, AI, and security:** User Management; Happy Birthday Trailhead 2023; Salesforce CRM; User Engagement; Lightning Experience Customization; Data Modeling; Agentforce 360 Platform Basics; Formulas and Validations; Correlation and Regression; Data Analytics Fundamentals; Natural Language Processing Basics; Artificial Intelligence Fundamentals; Generative AI Basics; Responsible Creation of Artificial Intelligence; Superbadge Program Security: Quick Look; Lightning Experience Basics; Data Management; CRM Analytics: Quick Look; Data Security; Reports & Dashboards for Lightning Experience.